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Optimizing Renewable Strategies for Emission Reduction Through Robotic Process Automation in Smart Grid Management

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Abstract: The integration of renewable energy into Intelligent Distribution Networks (IDNs) is challenged by the inherent variability and fluctuations in energy supply, particularly with photovoltaic (PV) generation as the primary form of distributed generation (DG). However, managing the fluctuations and variability in renewable energy supply presents significant challenges. To address these complexities, it is vital to optimally coordinate flexible resources from source–network–storage–load (SNSL) in a manner that aligns with cross-sectoral emission reduction strategies while enhancing grid stability and efficiency. This paper addresses these challenges by proposing a strategy that optimizes the coordination of PV-based DG, storage, and load resources through Robotic Process Automation (RPA) to enhance grid stability and support emissions reduction. We use a two-layer dispatching framework: the lower-layer model, formulated as a quadratic programming problem, maximizes PV utilization for individual users, while the upper-layer model, based on a second-order cone relaxation approach, manages the overall IDN to minimize operational costs. The iterative solution leverages tie-line power flow as boundary information to ensure convergence across the network. Validated on an enhanced IEEE 33-bus system, the approach demonstrates a 62% increase in PV-based DG consumption and a 25% reduction in active power losses, highlighting its potential to improve grid efficiency and contribute to emission reduction goals.

Keywords: emission; robotic process automation; distributed dispatching; resource management



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1. Introduction

The rapid advancement of smart grid technologies and the increasing integration of distributed generation (DG) into Intelligent Distribution Networks (IDNs) have introduced both opportunities and challenges in modern energy systems. Distributed energy sources, including solar, wind, and other renewable technologies, promise a more sustainable and decentralized energy future [1,2]. However, managing the inherent fluctuation and variability in energy supply poses significant challenges to grid stability, reliability, and economic efficiency [3,4]. The need to optimally coordinate flexible resources, such as source–network–storage–load (SNSL), in this complex landscape has become critical to ensure the seamless operation of smart grids [5]. In response, innovative technologies such as Robotic Process Automation (RPA) have gained prominence for their ability to automate repetitive tasks, streamline operations, and enhance the adaptability of grid management systems [6,7].

Previous research has extensively explored methods for optimizing distributed energy resources within Intelligent Distribution Networks. In [2], economic dispatching methods were examined with a focus on handling renewable energy uncertainties and demand-side responses, emphasizing the necessity for adaptive models to manage unpredictable

fluctuations in supply and demand. In [8], the study applies machine learning to optimize wireless charging for electric buses within a Nguyen–Dupuis network, focusing on enhancing efficiency and sustainability in urban transit. By modeling dynamic charging needs, the research offers insights into reducing infrastructure dependence and improving fleet performance through predictive charging strategies. In [9], the authors investigate dynamic wireless power transfer (DWPT) in a multi-depot, multi-product delivery supply chain, highlighting DWPT's potential to improve logistics efficiency. The study underscores DWPT's benefits in reducing downtime and infrastructure requirements, providing a framework for sustainable energy use in complex delivery networks. In [10], the authors propose an optimization planning method for distributed generation based on the steady-state security region of the distribution network. Both studies underline the importance of addressing variability in DG for improving the resilience and efficiency of distribution networks. Further advancements in optimizing microgrid clusters are presented by [11], who develop a distributed scheduling method based on the distributed Newton method. Ref. [12] focuses on the distributed multi-scenario optimal sizing of integrated electricity and gas systems, addressing the need for coordination across multiple layers of resource management. Ref. [13] proposes a distributed optimization method with weighted gradients for the economic dispatch of multi-microgrid systems, emphasizing the benefits of weighted optimization in complex energy systems. Ref. [14] introduces a genetic algorithm applied to optimize voltage-led customer load services, further illustrating the role of optimization algorithms in improving energy dispatching efficiency. Ref. [15] proposes a multi-criteria decision-making approach for the design and management of renewable energy communities, focusing on the integration of diverse renewable resources and energy storage solutions. The study emphasizes the need for system optimization and demand-side management to ensure that renewable energy is maximized while minimizing grid dependency and operational costs.

Managing uncertainty and variability in renewable energy sources is a recurring theme in the literature. Ref. [4] further expanded on the planning and management of distributed generations and energy storage using a bi-level programming approach, emphasizing the need for coordinated planning in active distribution networks. Ref. [16] addressed uncertainty in renewable energy generation and proposed methods for improving grid stability and dispatching. Ref. [17] introduced a stochastic economic dispatch method, considering voltage-sensitive loads as reserves, which improved flexibility in handling unpredictable changes in load demand. Ref. [18] addressed the challenges of deep photovoltaic (PV) penetration in distribution networks, where correlated net bus load forecast errors significantly affect system stability. Ref. [19] developed a fast sensitivity method for determining line loss and node voltages in active distribution networks. Ref. [20] proposed a two-layer, two-stage dispatching strategy for active distribution networks with microgrid interactions, highlighting the need for advanced optimization techniques that can handle the dynamic interactions between distributed resources. Ref. [21], a study on the integration of Soft Open Points (SOPs) for supply restoration in active distribution networks, offered a direct parallel to focus on optimizing power flows in IDNs. The SOP integration methods used here could enhance the approach by further improving flexibility and energy distribution in a proposed RPA-based dispatching model. Ref. [22] introduced advanced optimization techniques for energy storage systems using a dual-stage approach, which can complement the method used for integrating energy storage in IDNs. The use of improved gravitational search algorithms adds a novel dimension to energy management strategies that could inform enhancements to the optimization framework. Ref. [23] focused on large-scale electric vehicle (EV) dispatching through second-order cone programming, which resonates with the use of similar mathematical methods in a smart grid model. This paper's strategies for managing large-scale energy dispatch could offer new insights for extending a model to accommodate emerging technologies like EVs. Ref. [24] emphasized the optimization of battery energy storage systems for peak load shifting and voltage distribution, which aligns with work on improving energy storage management. The insights on peak load

management could further refine the dispatching strategy to handle load fluctuations more effectively in smart grids. Ref. [25] proposed decomposition and coordination methods for economic dispatch in active distribution networks, providing a strong foundation for the multi-layered approach proposed in their model. The paper's techniques for coordinating multiple microgrids could enhance the efficiency of the IDN framework, improving resource allocation across the network. Ref. [26], a study focusing on optimizing photovoltaic and energy storage integration in rural microgrids, offered useful parallels for work on distributed generation management. While the context is rural microgrids, the strategies for enhancing energy storage dispatch and renewable integration could be adapted to improve resource utilization in the IDN model. Ref. [27] introduced a two-stage demand response model with battery energy storage, providing valuable techniques for managing energy storage in a smart grid system. The optimization strategies for demand response could inform improvements in the approach to balancing active and reactive power in distributed networks.

Despite the extensive body of literature on smart grid optimization and automation technologies, a notable gap exists in the integration of RPA into the distributed dispatching and adaptive resource management of IDNs. While several studies have examined optimization techniques for dispatching DG and microgrid systems, the incorporation of RPA as a strategic tool for managing the complexities of SNSL resources remains underexplored. Existing models often lack the ability to fully automate resource coordination across multiple layers of a distribution network, resulting in suboptimal dispatching and delayed response to real-time fluctuations in energy supply. This study addresses key gaps in smart grid optimization by integrating Robotic Process Automation (RPA) into distributed dispatching and adaptive resource management within Intelligent Distribution Networks (IDNs), enabling automated and efficient real-time responses to energy fluctuations. By introducing a novel laminar flow structure inspired by communication systems, the research optimally coordinates source–network–storage–load (SNSL) resources, achieving streamlined pre-scheduled dispatching and enhanced grid stability. A unique two-layer dispatching model is developed, with a lower-layer quadratic programming model for distributed generation (DG) users and an upper-layer second-order cone relaxation model for the overall IDN. These models are solved iteratively, maximizing user economic benefits while minimizing operational costs for the IDN, thus enhancing flexibility and efficiency across network resources. Validated on the IEEE 33-bus system, this approach demonstrates significant improvements, including a 62% increase in DG consumption and a 25% reduction in active power losses, highlighting the potential of RPA-driven solutions in future smart grid management.

This paper is organized as follows: It begins with an Introduction Section that highlights the motivation for integrating Robotic Process Automation (RPA) into Intelligent Distribution Networks (IDNs) to tackle challenges in renewable energy management. The Related Work Section reviews existing research on smart grid optimization, coordinated dispatching models, and RPA applications, pinpointing a significant gap in automated resource management within IDNs. The Proposed Model introduces a two-layer dispatching framework, where a lower-layer quadratic programming model manages distributed generation (DG) resources and an upper-layer model, utilizing second-order cone relaxation, optimizes the entire IDN. The Results and Discussion Sections evaluate the model's effectiveness through simulations, including pre-scheduled dispatching strategies, sensitivity analyses on factors like DG output, and the influence of centralized resources on IDN performance, emphasizing improvements in DG utilization and reductions in power losses. Finally, the Conclusion Section summarizes this study's contributions, illustrating the potential of RPA-enhanced smart grid automation to improve flexibility and efficiency and proposing directions for future research in automated IDN management.

2. Related Work

2.1. Robotic Process Automation (RPA)

The use of Robotic Process Automation (RPA) in energy systems is an emerging field. RPA is recognized for its ability to automate repetitive tasks and streamline operations in various industries. Ref. [3] proposed a distributionally robust reactive power dispatch model, incorporating a Wasserstein distance to handle uncertainties in active distribution networks. Both studies contribute to the development of multi-layered optimization frameworks that align with the goals of this paper, particularly the integration of RPA into distributed dispatching. Ref. [7] demonstrated the development of intelligent RPA in a Brazilian utility case study, where automation enhanced the scalability and coordination of energy dispatching. Refs. [28,29] explored the impact of RPA on enterprise resource planning (ERP) systems and risk management in automation, providing insights into how automation can be applied in complex systems such as smart grids. Ref. [30] provided valuable insights into the security challenges surrounding Robotic Process Automation (RPA), which is critical for ensuring the reliability and safety of automated processes in smart grids. Although the focus is primarily on security, these insights are useful for strengthening the security framework in RPA-based energy dispatching system, ensuring a robust and secure operation. Ref. [31], a study on network reconfiguration and voltage regulation, offered relevant insights for managing unbalanced distribution systems, which could complement RPA-based dispatching in IDNs. The combination of reconfiguration and distributed energy resource control aligns well with goal of improving grid flexibility and stability. Ref. [32] introduced self-learning capabilities for RPA, which could significantly enhance the adaptability of the proposed RPA-based dispatching system. Although their primary application is in business processes, the self-learning mechanisms could be leveraged to improve real-time decision-making and automation in smart grids. Ref. [33] provided technical insights into the design and implementation of RPA systems, which are crucial for understanding how to optimize a RPA-based smart grid dispatching system. The system design strategies outlined here could help to improve the scalability and efficiency of RPA in the model. In [34], a study on the benefits of RPA in supply chains, the authors highlighted parallels with work in energy systems, emphasizing RPA's role in improving process efficiency. The benefit realization techniques discussed could inform how researchers measure and evaluate the impact of RPA on energy dispatching in a smart grid model. Ref. [35] covered RPA tools and technologies across various industries, providing a broad view of their applications, including in energy management. The exploration of RPA's industrial applications offers useful insights into how these tools can be adapted and scaled for energy grid management and optimization. Ref. [36] discussed the combination of RPA and AI-powered services to increase productivity and reduce waste, which could inspire the future integration of AI into RPA systems for enhancing optimization. Although primarily focused on industrial services, the AI-driven productivity improvements could be applied to smart grid management for greater efficiency. Ref. [37] explored RPA's transformative power in Industry 4.0, offering insights into its potential to revolutionize energy grid management by automating complex tasks. The focus on business transformation through RPA could provide a framework for understanding how RPA can reshape distributed energy management and improve automation in IDNs.

2.2. Coordinated Dispatching Model

The integration of diverse flexible resources, each managed by different stakeholders within an Intelligent Distribution Network (IDN), is depicted in Figure 1. In this setup, users possess the authority to dispatch distributed generation (DG) and distributed energy storage (ES) systems. On the other hand, the Grid Management Operator (GMO) has control over the dispatching of centralized ES, capacitor banks (CBs), static var generators (SVGs), and other equipment owned by the IDN. The dispatching process for all these resources can be structured into a primary problem and several sub-problems. The primary

problem corresponds to the IDN's overall dispatching model, while the sub-problems represent the dispatching models for individual users with DG.

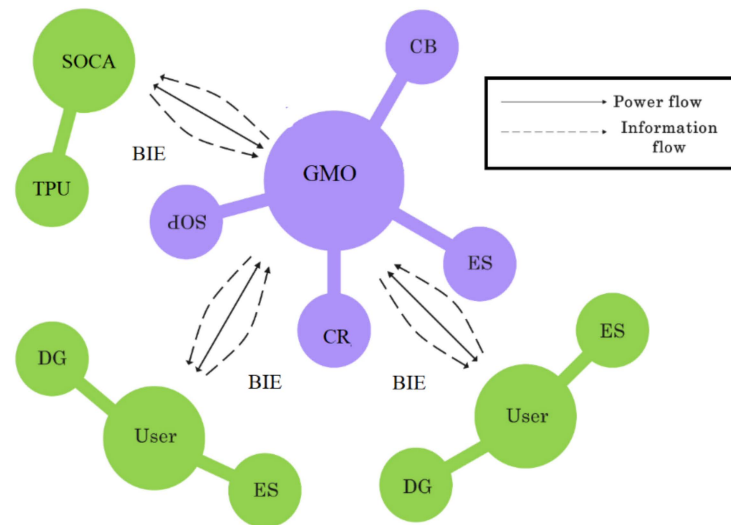


Figure 1. Integration of flexible resources and stakeholder roles within the Intelligent Distribution Network (IDN).

For users with DG, the optimization objective is to formulate a dispatching strategy for their DG and ES resources based on a TPU pricing mechanism to maximize economic benefits. Meanwhile, the Distribution System Operator (DSO) must consider the safe and reliable operation of the entire IDN. The power flow on the tie-line serves as the Boundary Information Exchange (BIE) between the upper-level model (IDN dispatching) and the lower-level model (user dispatching), facilitating coordination between the two levels. Ref. [38] explores state-of-the-art technologies in solar-powered DC microgrids with hybrid energy storage systems, emphasizing the roles of converter topologies in optimizing energy flow. They highlight that the integration of a hybrid approach addresses the challenges of variable renewable energy generation by improving system flexibility and enhancing power quality. Ref. [39] proposes a power management strategy for DC microgrids with a focus on state-of-charge (SOC) restoration and voltage stability through an improved voltage-shifting controller. Their approach optimizes energy storage by ensuring efficient charge–discharge cycles and maintaining a stable voltage, crucial for integrating renewable energy sources.

The dispatching model for users with DG focuses on optimizing the generation and consumption of energy from DG and distributed ES resources to achieve maximum economic benefits while maintaining grid stability. The optimization is conducted over a given time horizon, often divided into time slots (e.g., hourly intervals), based on a time-of-use (TPU) pricing mechanism that reflects the variation in electricity prices throughout the day.

The primary goal of the user dispatching model is to maximize economic benefits, which are achieved by optimizing the energy generated by DG, the power stored in or discharged from the ES system, and the energy exchanged with the grid. The optimization is impacted by TPU pricing, which determines when it is most cost-effective to sell or buy electricity.

The objective is to maximize revenue from selling excess energy generated by the DG to the grid while minimizing the cost of buying energy and storing it in the ES. The objective function can be mathematically expressed as

$$J_{Max.} = \sum_{t=1}^T [P_{DG}(t) \cdot \lambda_s(t) - PES_{in}(t) \cdot \lambda_b(t) - C_{op}(P_{DG}(t))] \quad (1)$$

where J is the total economic benefit over the time horizon T , $P_{DG}(t)$ is the power generated by DG at time t , $\lambda_s(t)$ is the electricity selling price at time t , $PES_{in}(t)$ is the power stored

in the ES system at time t (i.e., the energy purchased from the grid for storage in the ES system), $\lambda_b(t)$ is the electricity buying price at time t , and $C_{op}(P_{DG}(t))$ is the operational cost associated with running the DG at time t .

The dispatching model is subject to a set of operational and technical constraints that ensure the safe and efficient operation of both the user's energy system and the overall Intelligent Distribution Network (IDN).

At each time step, the total power generated, stored, and imported/exported must balance with the user's load demand. This is represented as

$$P_{DG}(t) + P_{ES_out}(t) + P_{grid}(t) = P_{load}(t) + P_{ES_in}(t) \quad (2)$$

where $P_{DG}(t)$ is the power generated by DG at time t , $P_{ES_out}(t)$ is the power discharged from the ES system at time t , $P_{grid}(t)$ is the power exchanged with the grid at time t (positive for import and negative for export), $P_{load}(t)$ is the user's total load demand at time t , and $P_{ES_in}(t)$ is the power charged into the ES system at time t .

This constraint ensures that the user's power generation, storage, and exchange with the grid meet their energy consumption requirements.

The energy storage system must operate within its capacity and power limits. The charging and discharging of the ES are governed by the following constraints:

$$P_{ES_min} \leq P_{ES_out}(t), P_{ES_in}(t) \leq P_{ES_max} \quad (3)$$

where P_{ES_min} and P_{ES_max} represent the minimum and maximum power that can be discharged from/charged into the ES system, respectively.

The state of charge (SOC) of the energy storage must be maintained within acceptable limits to prevent overcharging or deep discharging:

$$SOC_{min} \leq SOC(t) \leq SOC_{max} \quad (4)$$

The SOC of the ES at any time t is updated based on the charging and discharging operations as follows:

$$SOC(t+1) = SOC(t) + \eta_{ch} \cdot P_{ES_in}(t) \cdot \Delta t - \frac{P_{ES_out}(t)}{\eta_{dis}} \cdot \Delta t \quad (5)$$

where η_{ch} and η_{dis} are the charging and discharging efficiencies, respectively, while Δt is the duration of each time step.

The power output of the DG is limited by its operational capacity, ensuring it operates within safe limits:

$$P_{DG_min} \leq P_{DG}(t) \leq P_{DG_max} \quad (6)$$

where P_{DG_min} and P_{DG_max} represent the minimum and maximum power output capacities of the DG, respectively.

The time-of-use pricing mechanism affects the dispatching strategy. The user aims to purchase electricity when prices are low (e.g., during off-peak hours) and sell excess electricity when prices are high (e.g., during peak hours). The optimization problem seeks to adjust the timing of energy generation, consumption, and storage to take advantage of the varying electricity prices throughout the day: $\lambda_s(t) > \lambda_b(t)$ during peak hours and $\lambda_b(t) < \lambda_s(t)$ during off-peak hours. This TPU-based strategy ensures that energy is bought and sold at the most cost-effective times, leading to optimized economic benefits for the user.

In addition to meeting the user's economic objectives, the safe operation of the IDN must be ensured. This includes maintaining grid stability by ensuring that power flows on

the tie-line between the user and the grid remain within allowable limits, as well as that voltage and frequency stability are preserved:

$$P_{tie_min} \leq P_{tie}(t) \leq P_{tie_max} \quad (7)$$

where $P_{tie}(t)$ represents the power flow across the tie-line between the user's system and the IDN at time t , while P_{tie_max} defines the limits within which power flows must be maintained to ensure the safe operation of the IDN.

To ensure voltage stability in the IDN, the voltage at each bus must be maintained within predefined limits. The voltage at bus i at time t , denoted as $V_i(t)$, should remain within the allowable range:

$$V_{min} \leq V_i(t) \leq V_{max} \quad (8)$$

where V_{min} and V_{max} are the minimum and maximum allowable voltage levels, respectively, $V_i(t)$ is the voltage at bus i at time t .

The GMO can control voltage stability using devices such as capacitor banks (CBs) and static var generators (SVGs), which adjust reactive power flows to maintain acceptable voltage levels across the network.

3. Proposed Model

The optimization of both user and Intelligent Distribution Network (IDN) objectives within the RPA process is structured in a two-layer model. The lower-layer model aims to maximize the user's economic benefit by optimizing the use of distributed generation (DG) and energy storage (ES) systems. This is achieved by maximizing revenue from selling electricity, minimizing costs from buying power, and accounting for operational expenses of photovoltaic (PV) and ES systems over a given time period T . The objective function in this layer includes terms for selling and buying electricity prices, $\lambda_s(t)$ and $\lambda_b(t)$, and the total operating costs, $C_{total}(t)$, which depend on PV generation and ES charging efficiency. This lower-layer model is solved using quadratic programming, balancing power flows and SOC within specified limits. In the upper-layer model, the focus shifts to minimizing the IDN's total operational costs, incorporating expenses for grid power purchases, energy storage operations, and system losses, while maintaining voltage stability and power quality. The objective function includes terms for power exchange with the upper-level grid, system losses, and device operating costs, managed under a power flow balance constraint. The optimization process, involving SOPs, SVGs, CBs, and ES systems, uses an iterative approach where decisions made in the user layer feed into the IDN's cost minimization, ensuring that power supply meets demand. The IDN's model is structured as a least squares optimization, aiming to balance deviations from optimal power flow paths. This coordinated, iterative optimization, facilitated by RPA, effectively aligns the economic objectives of users with the stability and operational efficiency of the IDN, achieving robust energy management across the smart grid.

The RPA process for the user dispatching model is designed to maximize the user's net economic benefit by considering revenue from selling electricity, costs from buying electricity, and operational costs associated with the PV and ES systems.

For both the user and the distribution network, RPA automates the maximization or minimization of objective functions (economic benefits for the user and operational costs for the IDN).

User objective (lower-layer model):

$$\text{Maximize } J_{user} = \sum_{t=1}^T \sum_{t=1}^T \left[\lambda_s(t) \cdot P_{sell}(t) - \lambda_b(t) \cdot P_{buy}(t) - C_{total}(t) \right] \quad (9)$$

where $C_{total}(t)$ includes the cost of operating renewable energy resources and energy storage systems.

IDN Objective (upper-layer model):

$$\text{Minimize } JSDN = \sum_{t=1}^T [\lambda_p(t) \cdot P_{grid}(t) + CES(t) + C_{loss}(t) + Cop(t)] \quad (10)$$

where J is the total economic benefit over the time horizon T , $\lambda_s(t)$ is the selling price of electricity at time t , $P_{sell}(t)$ is the power sold to the grid at time t , $\lambda_b(t)$ is the buying price of electricity at time t , and $P_{buy}(t)$ is the power purchased from the grid at time t ;

$C_{total}(t)$ is the total cost function, which includes the costs associated with operating the PV and ES systems:

$$C_{total}(t) = (\alpha_{PV} + \beta_{PV}) \cdot P_{solar}(t) + \lambda_b(t) \cdot P_{ES_in}(t) \cdot \frac{1}{\eta_{ch}} + (\alpha_{ES} + \beta_{ES}) \cdot P_{ES_out}(t) \quad (11)$$

where α_{PV} and β_{PV} are the maintenance and degradation costs per unit of PV power generated, $P_{solar}(t)$ is the power generated by the PV system at time t , η_{ch} is the charging efficiency of the ES system, and α_{ES} and β_{ES} are the maintenance and degradation costs per unit of power discharged from the ES system.

The RPA transformation for the Intelligent Distribution Network (IDN) dispatching model automates the coordination of resources across the network. This includes centralized control of grid assets such as capacitor banks (CBs), static var generators (SVG), and centralized energy storage. RPA automates the process of balancing demand and supply while ensuring voltage stability and power quality across the network.

IDN objective (upper-layer model):

$$\text{Minimize } C_{IDN} = \sum_{t=1}^T [\lambda_p(t) \cdot P_{grid}(t) + C_{ES}(t) + C_{loss}(t) + C_{op}(t)] \quad (12)$$

where $\lambda_p(t)$ is the cost of purchasing power from the upper-level grid, $P_{grid}(t)$ is the power imported from the upper-level grid, $C_{ES}(t)$ is the cost of operating centralized energy storage, $C_{loss}(t)$ is the system losses, and $C_{op}(t)$ is the operating cost of devices such as CBs and SVGs.

The RPA automates power flow balance for the IDN:

$$P_{grid}(t) + P_{ES_out}(t) + \sum_{i=1}^N P_{DG_i}(t) = P_{load}(t) + P_{ES_in}(t) + P_{loss}(t) \quad (13)$$

where $P_{grid}(t)$ represents the power exchanged between the IDN and the transmission network, $P_{ES_out}(t)$ and P_{ES_in} represent the charging and discharging of the centralized ESS, and $P_{DG_i}(t)$ represents the power from distributed generation units within the network.

RPA applies optimization algorithms such as quadratic programming to solve the dispatching problems. These algorithms ensure that the objectives are met under the given constraints. The iterative optimization can be expressed as follows:

- Lower-layer model (user):

$$\text{Minimize } f_{user}(x) = \frac{1}{2}x^T Qx + c^T x \quad \text{subject to } Ax \leq b \quad (14)$$

where x represents decision variables (such as power flows, charging/discharging rates, etc.), and Q , c , A , and b represent matrices/vectors defining the problem structure.

Since the objective function is a quadratic form with a positive semi-definite matrix Q , this formulation ensures convexity. This convex nature guarantees that the solution, when it exists, will be globally optimal, making the lower-layer model an efficient QP problem.

- Upper-layer model (SDN):

$$\text{Minimize } f_{SDN}(y) = \|P_{grid} - P_{demand}\|^2 \quad (15)$$

where y represents the decision variables for the IDN model, and the cost function minimizes deviations from the optimal power flow.

Since this function minimizes the squared deviation from optimal power balance, it is convex and, more specifically, a linear programming (LP) problem due to the linear relationship between supply and demand.

The iterative optimization method ensures that both the user and IDN objectives align by iteratively solving these convex optimization problems. The convexity of each layer (user's QP and IDN's LP) ensures computational efficiency and convergence to a globally optimal solution.

The solving process for both the user and IDN dispatching models involves an iterative optimization technique that seeks to find a solution that satisfies the objectives and constraints of both models, as shown in Figure 2. Since the two systems are interconnected, the process must account for the boundary exchange of power between the user and the IDN. In this model, Equations (9) and (14) in the lower-layer (user) model and Equations (10) and (15) in the upper-layer (IDN) model are interconnected as they collectively work toward optimizing energy usage and minimizing costs within the Intelligent Distribution Network (IDN). Each pair serves a distinct but complementary purpose in terms of achieving economic efficiency and reducing carbon emissions, with the iterative optimization ensuring alignment between individual user benefits and overall network stability. In essence, Equation (9) defines the target economic objectives for the user, while Equation (14) provides the means to mathematically optimize the decisions needed to reach those objectives within the practical constraints of the energy system. In addition, Equation (10) defines the network's economic and operational objectives, while Equation (15) provides the mathematical optimization framework required to achieve these objectives by balancing supply and demand across the network.

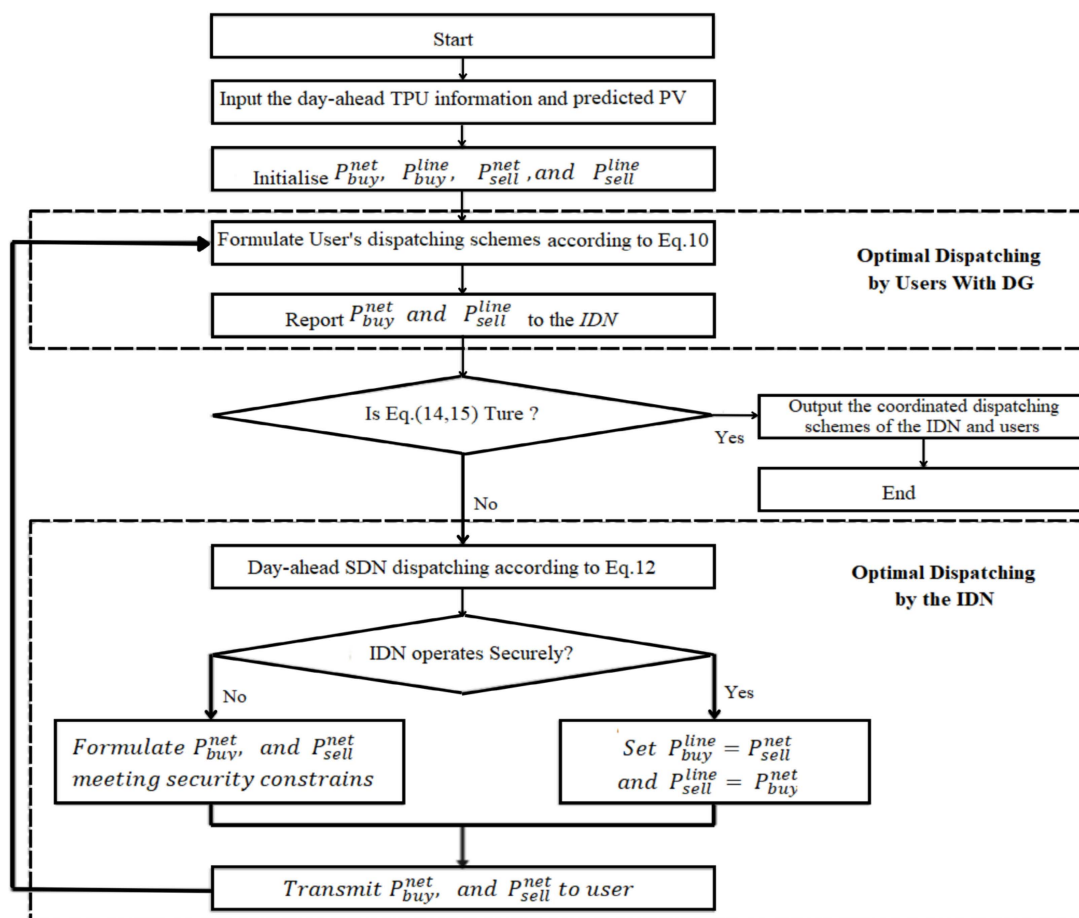


Figure 2. Iterative solution process for distributed dispatching models in the user and IDN systems.

The solution process involves several steps:

Step 1: Initialize all variables, including $SOC(t)$, $P_{solar}(t)$, $P_{wind}(t)$, and forecasts for $P_{load}(t)$.

Step 2: The RPA system solves the user dispatching model based on the objective of maximizing user benefit by following Equation (10).

Step 3: The IDN dispatching model is solved by the RPA system to minimize operational costs in Equation (12).

Step 4: The RPA system iterates through different dispatch strategies to minimize costs and ensure system stability.

Step 5: The power exchanged on the tie-line between the user and the IDN serves as boundary information in Equation (7).

Step 6: The RPA checks for convergence. If the dispatching models for both the user and the IDN converge (i.e., the power balance and all constraints are satisfied), the system proceeds to the next time step. If not, the process is repeated.

Step 7: The entire process is repeated for each time step $t = 1, 2, \dots, T$, ensuring that both the user and the IDN models are solved iteratively and optimally.

4. Result and Discussion

4.1. Simulation Parameters

This detailed configuration of the IEEE 32-bus system, with its combination of SOPs, DG, ES, SVG, and CBs, creates a robust simulation environment that accurately reflects the challenges and dynamics of modern smart grid management, enabling a thorough evaluation of optimization and dispatching strategies in a real-world-like scenario, as shown in Figure 3. Key parameters include SOP power exchange limits, control settings, and voltage regulation ranges to balance power and maintain stability. DG units are characterized by variable generation capacities (0.5 to 2 MW) and intermittency factors that represent renewable variability, while ES systems have defined state-of-charge (SOC) limits (20% to 90%), charging/discharging efficiencies (85% to 95%), and degradation rates for realistic energy storage modeling. SVGs provide reactive power support within ± 0.5 MVar to stabilize voltage, with fast response times required to adjust to load fluctuations, and CBs offer fault isolation with rated interrupting capacities in kiloamperes (kA) to enhance network protection.

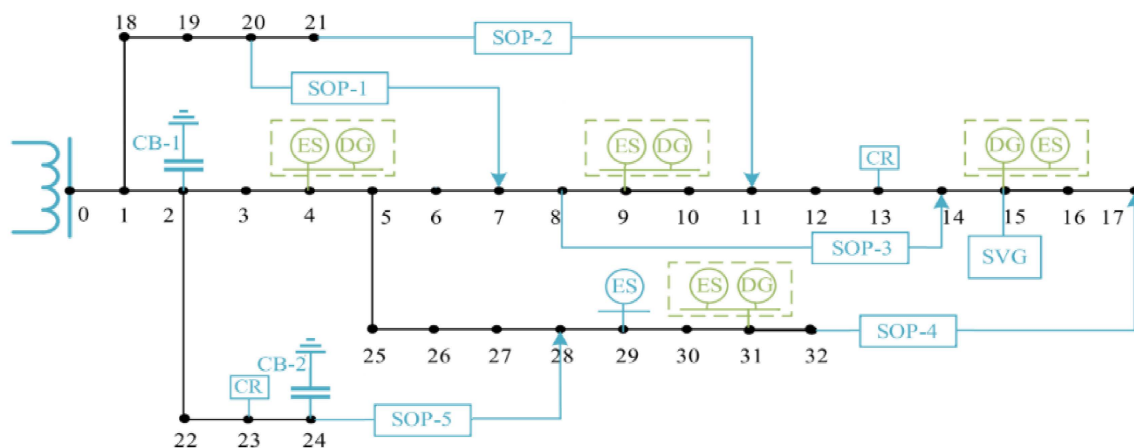


Figure 3. The configuration of the IEEE 32-bus testing system for smart grid simulation.

Figure 4 presents the typical daily power variations for two distinct seasons, namely summer and winter, showcasing the differences in both load demand and photovoltaic (PV) power generation. In Figure 4a, the demand load profiles for summer and winter are plotted over a 24 h period. The graph demonstrates that the summer load profile remains relatively stable during the early hours of the day, with a slight increase starting at 6 AM and reaching its peak between 12 PM and 8 PM. In contrast, the winter load profile shows

a more pronounced increase in the early hours of the morning, reaching a higher peak compared to summer, and then maintaining a steady level throughout the afternoon before gradually declining in the evening. Figure 4b depicts the PV output for both seasons. In winter, the PV output begins earlier, i.e., around 8 AM, and reaches its maximum output between 12 PM and 1 PM, before tapering off towards the evening. During summer, the PV output begins slightly later, i.e., around 9 AM, but the peak output is sustained for a longer period between 11 AM and 4 PM, showcasing the higher potential for solar energy generation during the summer months.

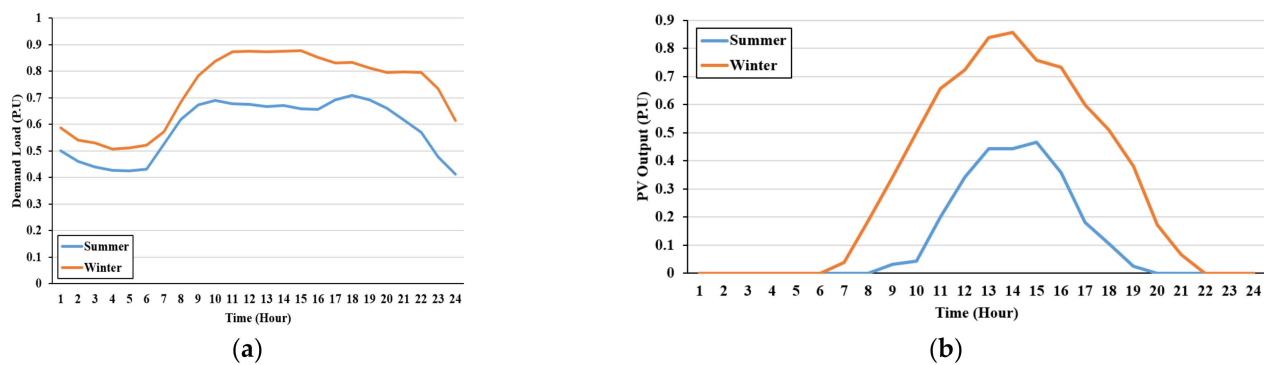


Figure 4. Typical daily power variations for two seasons: (a) load demand profiles; (b) PV power generation.

This comparison highlights the seasonal variation in both load demand and PV power generation, emphasizing the need for adaptive strategies in smart grid optimization to effectively manage the fluctuations in both supply and demand across different seasons.

Figure 5 provides a detailed breakdown of electricity tariffs per unit (TPU) over a 24 h period, focusing on the differences between selling and purchasing electricity. The TPU selling rates, which refer to the price at which electricity can be sold back to the grid, generally fluctuate throughout the day, with significant increases in the late morning and early afternoon. Specifically, the highest selling tariffs are observed between 11 AM and 1 PM, making these hours optimal for selling electricity generated by distributed generation units or other sources of excess power. Outside of these peak hours, the selling tariffs gradually decrease, reaching their lowest points during late-evening and early-morning hours.

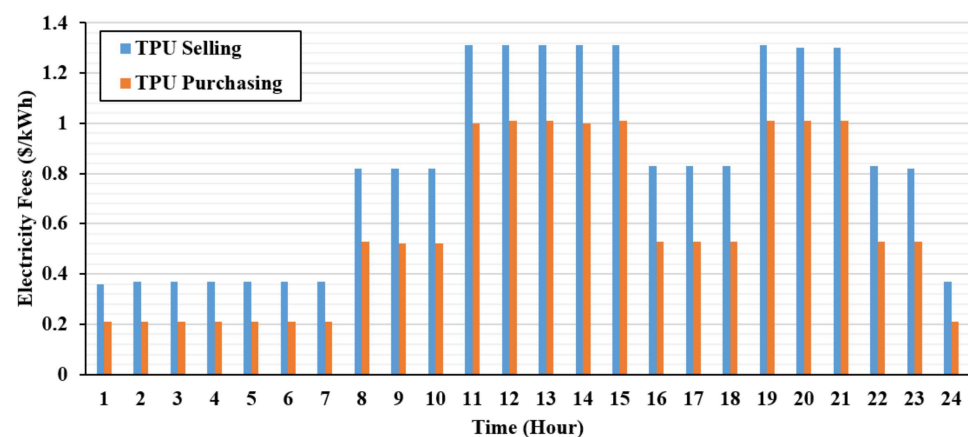


Figure 5. Electricity tariffs per unit (TPU) for selling and purchasing over a 24-h period.

4.2. Evaluation of the Pre-Scheduled Dispatching Strategy

(1) Preliminary Dispatching Plans for Users

Figure 6 provides an overview of the initial dispatching plans for Distributed Generation (DG) units at four different buses: Bus 5, Bus 9, Bus 15, and Bus 31. Key parameters

illustrated include purchasing power, selling power, photovoltaic (PV) power generation, and the states of charge (SOCs) of energy storage systems over a 24 h period. In Figure 6a for Bus 5, PV power generation begins to increase in the morning, reaching its peak at around 12 PM with approximately 5500 kW, leading to a significant rise in SOC, which reaches nearly 100% capacity by midday. During non-PV hours, particularly in the early morning (1–5 AM) and late evening (8–10 PM), purchasing power is at its highest, reaching levels close to 5000 kW, while selling power remains minimal throughout the day, indicating local consumption or storage of the generated energy. In Figure 6b at Bus 9, a similar trend is observed. PV generation peaks around noon with a maximum output of approximately 3800 kW, which contributes to a high SOC nearing 90% during the middle of the day. This stored energy reduces the need for purchasing power from the grid, particularly during peak PV hours, keeping the purchasing power demand relatively low during the afternoon. The selling power is minimal, indicating that most of the energy generated is used locally or stored for later use. In Figure 6c representing Bus 15, the dispatching pattern mirrors those of the other buses, with PV generation peaking at around midday at approximately 4200 kW. The SOC steadily rises, reaching a peak around midday. Purchasing power remains steady during non-PV hours, with peaks occurring in the early morning and late evening. Similar to the other buses, the selling power at Bus 15 is low, indicating that the DG output is primarily used for local demand or stored. Finally, Figure 6d for Bus 31 displays trends consistent with the previous buses. PV generation reaches its maximum of around 3500 kW at around midday, which leads to a substantial increase in SOC. Purchasing power is higher during non-PV periods, particularly in the early hours (1–5 AM) and later in the evening (8–10 PM), reaching up to 4000 kW. Selling power remains consistently low, suggesting that the generated energy is utilized locally or stored.

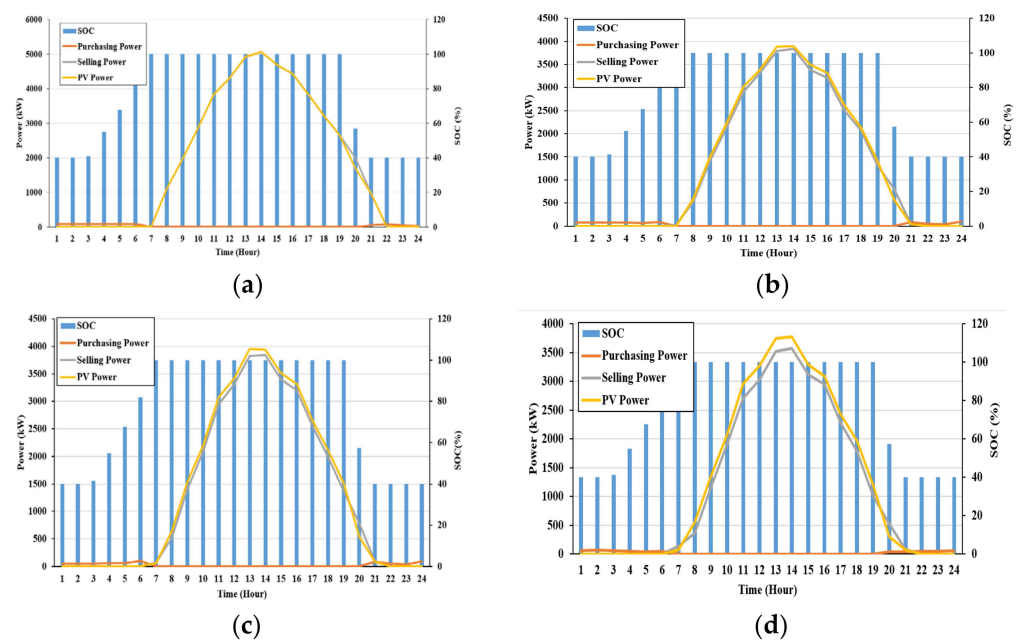


Figure 6. Initial dispatch plans for Distributed Generation (DG) units at (a) Bus 5, (b) Bus 9, (c) Bus 15, and (d) Bus 31.

To ensure operational resilience, a minimum SOC threshold of 40% is maintained across all buses, particularly to prevent deep discharges that could impact battery longevity and reliability. This constraint is especially relevant during non-PV hours, such as in the early morning and late evening when grid power is purchased to maintain SOC at or above 40%. By enforcing this SOC limit, the system reduces the need for discharging below safe levels, resulting in a more sustainable and resilient battery operation. This 40% SOC threshold also limits selling power during low-SOC periods, prioritizing storage and local consumption over grid export to maintain a stable reserve of energy. This approach

enhances the stability of energy storage systems across all buses, aligning the dispatching strategy with practical and sustainable operational requirements.

(2) Impact of PV Consumption

Figure 7 illustrates the impact of PV consumption on the voltage profiles across all buses in the Intelligent Distribution Network (IDN), highlighting how distributed generation (DG) from PV systems affects grid stability throughout the day. As PV generation increases during daylight hours, particularly around midday, the voltage levels rise significantly, reaching values above 1.2 P.U. at certain buses. This not only reflects the ability of PV systems to enhance local voltage levels but also underscores the importance of maintaining voltage within safe operational limits to prevent overvoltage scenarios. Figure 7 shows a noticeable fluctuation in voltage throughout the day, with lower values during early morning and late evening when PV generation is minimal and higher values during peak generation hours.

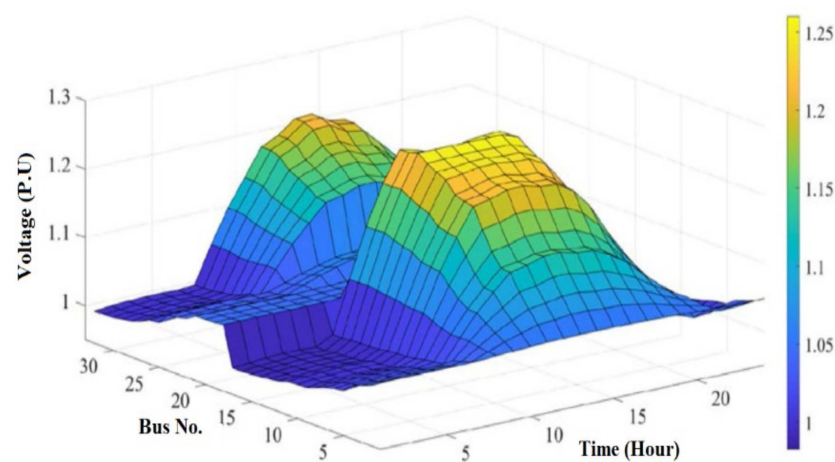


Figure 7. Voltage profiles across all buses with distributed generation (DG) operation.

(3) User Dispatching Scheme with Consideration of BIE

Figure 8 illustrates the final dispatching strategy for distributed generation (DG) units at four buses—Bus 5, Bus 9, Bus 15, and Bus 31—incorporating the Benefit of Integrated Energy (BIE) into the Intelligent Distribution Network (IDN). In Figure 8a for Bus 5, energy storage exhibits a significant increase in the state of charge (SOC) at midday as PV generation peaks at around 3500 kW, efficiently utilizing the surplus solar energy. The SOC reaches approximately 90% at midday, reducing the necessity to purchase power from the grid, which is seen at lower values (around 500 kW) during these hours. Minimal energy, approximately 200 kW, is sold back to the grid, as most of the generated energy is either stored or used locally. In Figure 8b at Bus 9, a similar energy management pattern is observed, with the SOC increasing significantly during midday. PV generation peaks at approximately 2200 kW, boosting the SOC to nearly 85%, enabling the storage system to absorb excess energy. This decreases the demand for purchasing power from the grid to below 300 kW. Selling power remains minimal, i.e., around 150 kW, indicating that generated energy is primarily stored or consumed on-site. Figure 8c, representing Bus 15, displays a more dynamic profile with two SOC peaks: one in the morning at around 8 AM, corresponding to a PV generation of 1200 kW, and another at around 2 PM, with a PV output reaching approximately 1500 kW. During these periods, the SOC rises to around 70%. Purchasing power from the grid remains steady, with slight increases during early-morning and late-evening hours when PV generation is low. Selling power also shows a slight increase, reaching around 200 kW in the afternoon, indicating that some excess energy is exported back to the grid. In Figure 8d for Bus 31, the energy storage system reaches its highest SOC of nearly 100% at around midday, corresponding to peak PV generation of about 2000 kW. The grid purchasing power is highest during the early

morning and late evening (up to 1500 kW), when solar generation is minimal. Selling power stays low, averaging around 100 kW throughout the day, highlighting a focus on local energy consumption and storage.

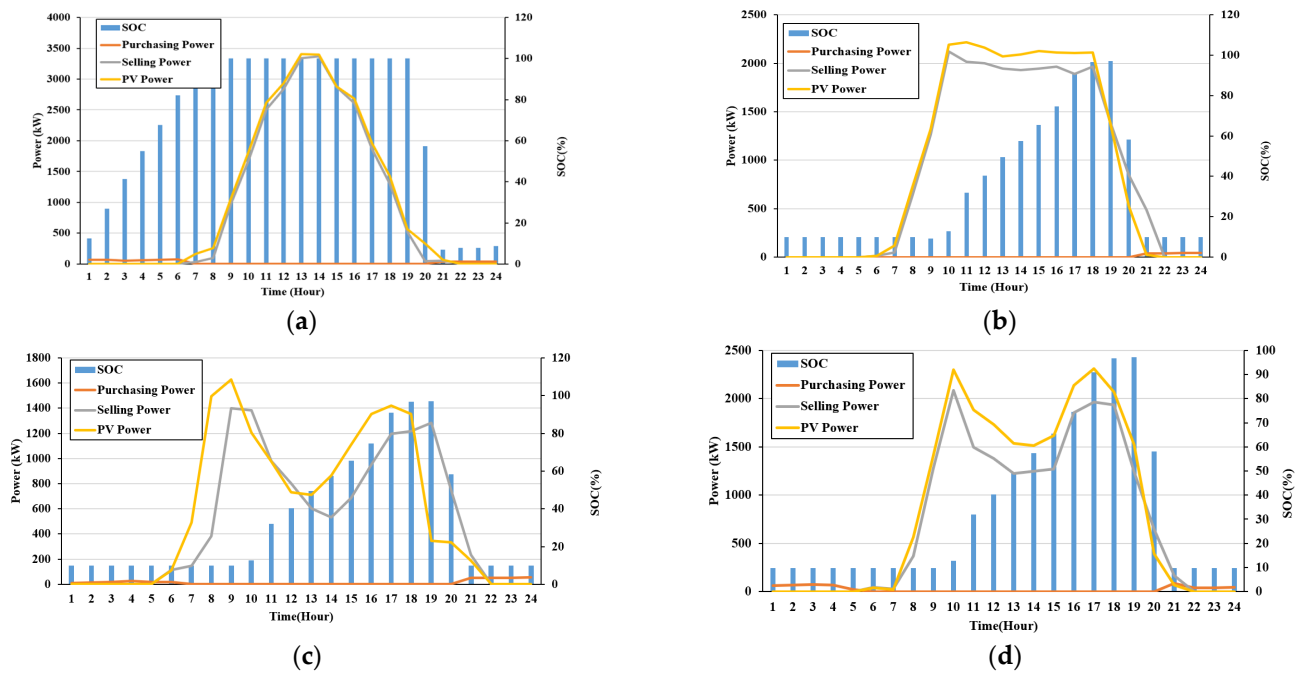


Figure 8. Final dispatching strategy for distributed generation (DG) units at (a) Bus 5, (b) Bus 9, (c) Bus 15, and (d) Bus 31, incorporating the Benefit of Integrated Energy (BIE).

Overall, Figure 8 demonstrates the effectiveness of the BIE strategy at optimizing DG and ES operations. It reduces grid dependency during peak solar hours by storing energy when PV generation is high and using stored energy during low-generation periods. This approach not only ensures efficient energy utilization but also minimizes energy purchases from the grid while maintaining local energy stability across the network.

(4) IDN Dispatching Strategy with Modified BIE Considerations

Figure 9 presents the voltage profile across all buses in the Intelligent Distribution Network (IDN) while considering the dispatching strategy with modified Benefit of Integrated Energy (BIE) adjustments. The voltage levels at each bus are shown over time, where the modified BIE has been designed to optimize energy consumption, generation, and storage while ensuring that safety and operational constraints are met. The figure shows a relatively stable voltage profile, with minor variations across the buses throughout the day. Most of the voltage values remain near the nominal level of 1.0 P.U., indicating that the dispatching strategy successfully maintains voltage stability while integrating distributed generation (DG) and PV power systems. The slight dips and peaks in the voltage profile across various buses represent the dynamic adjustments made to account for power generation from distributed sources and the load variations over time. The voltage levels slightly exceed 1.02 P.U. at certain times, while at others, they drop just below 1.0 P.U., reflecting the effects of real-time energy management strategies and demand response actions based on the modified BIE. The network remains well within safe operating limits, ensuring reliable grid performance even as energy demand and generation fluctuate.

Figure 10 illustrates the power exchange dynamics and the corresponding bus voltage throughout a 24 h period in the Intelligent Distribution Network (IDN). The figure provides a clear depiction of how purchasing and selling power from the grid impacts the voltage at the root node over the course of the day. In the first few hours, purchasing power remains relatively high, corresponding to lower voltage levels, indicating a period of grid dependency when local distributed generation (such as PV) is not active. As the

day progresses, purchasing power gradually decreases, coinciding with a rise in selling power, particularly around midday. During this time, the voltage stabilizes around the nominal value, reflecting the effect of efficient energy management and exchange strategies. The relationship between power exchanges—both purchasing and selling—and the voltage profile highlights the IDN's ability to balance energy demand and generation while maintaining voltage stability across the network.

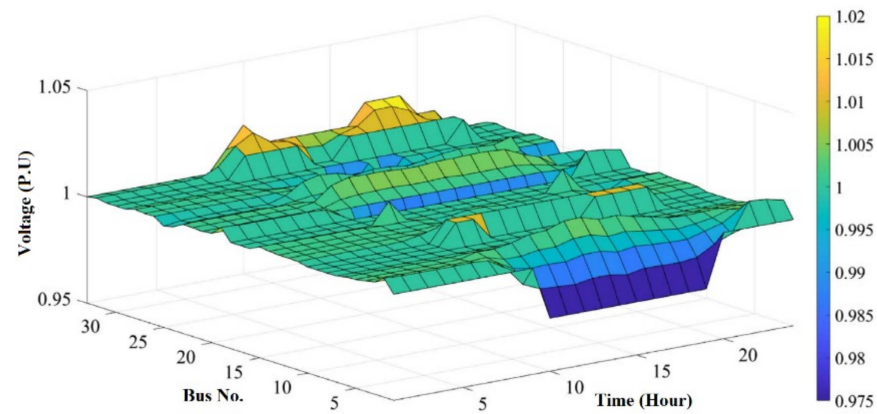


Figure 9. Voltage profiles across buses with a dispatch strategy incorporating the modified Benefit of Integrated Energy (BIE).

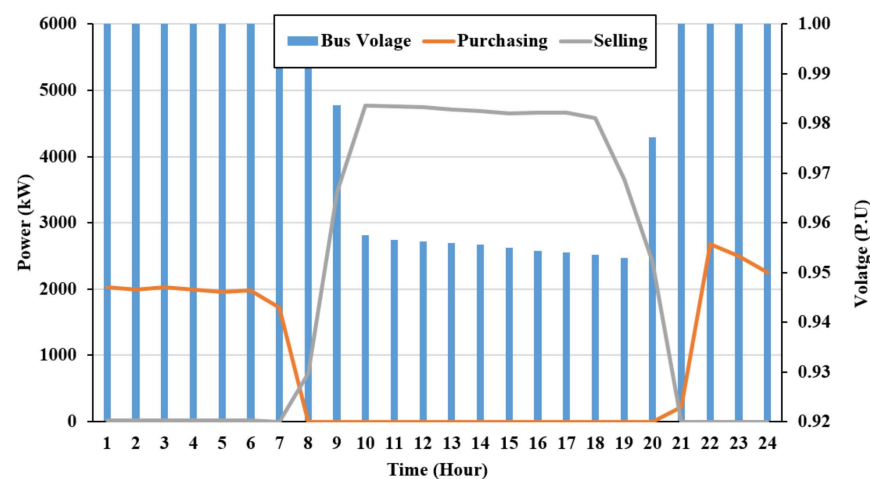


Figure 10. Power exchange dynamics and root node voltage profile in the Intelligent Distribution Network (IDN).

Figure 11 demonstrates the influence of key equipment, including Circuit Breakers (CB-1 and CB-2), the static var generator (SVG), and energy storage (ES), on the management of reactive power in the Intelligent Distribution Network (IDN) over a 24-h period. The circuit breakers (CB-1 and CB-2) operate consistently, ensuring network protection and stability by isolating sections of the grid when necessary. The SVG plays a pivotal role in dynamically regulating reactive power, adjusting its output to maintain voltage stability and improve power quality, especially during periods of fluctuating load or generation. Energy storage (ES) systems contribute significantly by managing both active and reactive power, providing reactive power support during critical times, particularly when demand peaks or generation from distributed sources varies. Together, these components work in tandem to ensure that reactive power levels are effectively controlled, maintaining voltage stability and ensuring the overall reliability of the grid.

Figure 12 illustrates the impact of Soft Open Points (SOPs) on power flow across different buses in the Intelligent Distribution Network (IDN) over a 24 h period. The

graph shows the activities of SOP-1, SOP-2, SOP-3, and SOP-5, highlighting their roles in balancing power between buses by either absorbing or supplying energy based on network demand and generation conditions. During early hours, SOP-1 and SOP-2 are primarily absorbing power, as reflected by the negative values, helping to alleviate grid congestion during periods of off-peak demand. As the day progresses, the need for SOP intervention decreases due to higher local generation, especially from distributed sources like photovoltaic systems. In the late afternoon and evening, all SOPs, particularly SOP-2, shift to supplying power, helping to meet increased demand as local generation declines. The dynamic behavior of these SOPs demonstrates their critical role in enhancing grid flexibility, maintaining voltage stability, and optimizing power flow by redistributing energy between buses during both high-generation and high-demand periods.

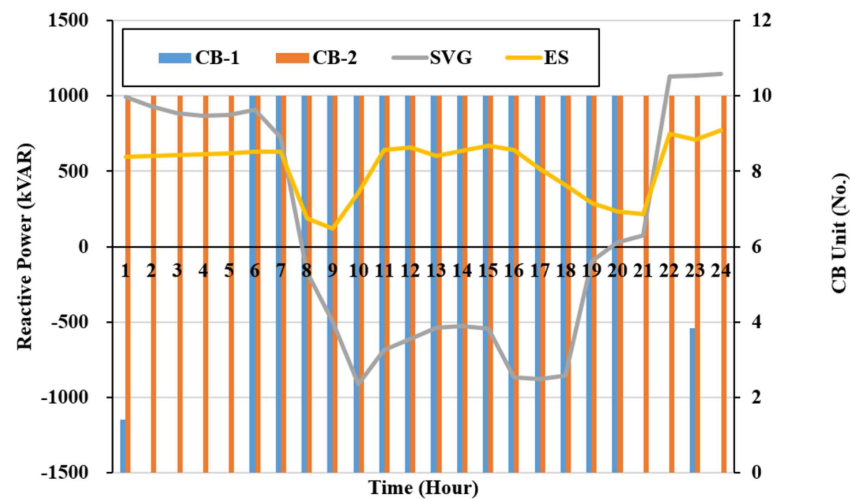


Figure 11. Impacts of circuit breakers, static var generator (SVG), and energy storage (ES) on reactive power management in the IDN.

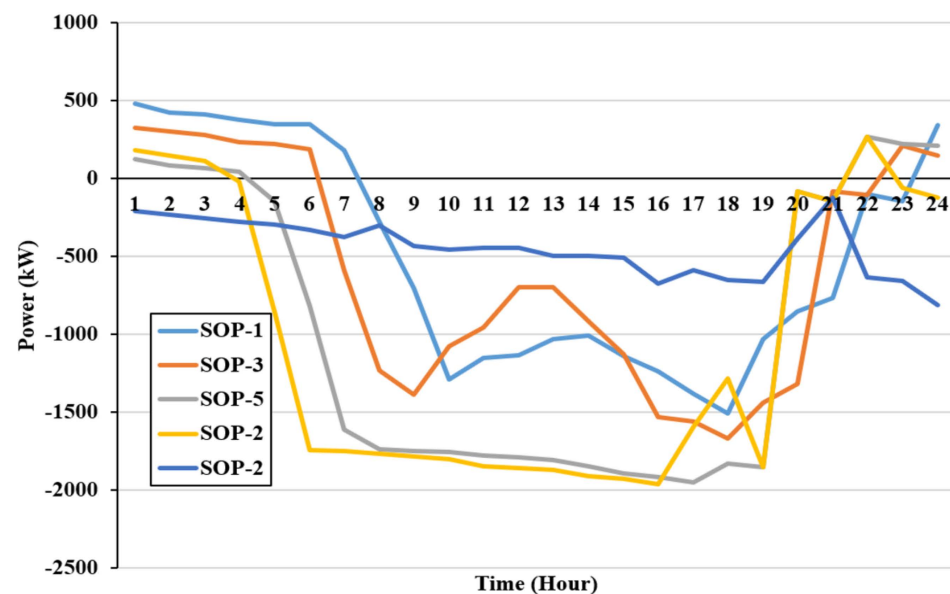


Figure 12. Impact of Soft Open Points (SOPs) on power flow across buses in the Intelligent Distribution Network (IDN).

Figure 13 demonstrates the impact of Compensation Ratio (CR) optimization on the load power at Bus 23 and Bus 6 over a 24 h period. The figure compares the original load power profiles with those optimized using CR strategies. For Bus 23, the original load shows a gradual increase in demand during the day, peaking in the afternoon. After

optimization, the bus can handle higher demand more efficiently, resulting in a more pronounced and smoother peak, reflecting improved resource utilization. For Bus 6, which has lower overall demand, the optimization leads to a slight increase in load power during peak hours, helping to better distribute the load and reduce system stress. In both buses, the optimized CR strategy enhances the system's ability to manage fluctuating demand, leading to a more balanced and stable load profile across the network.

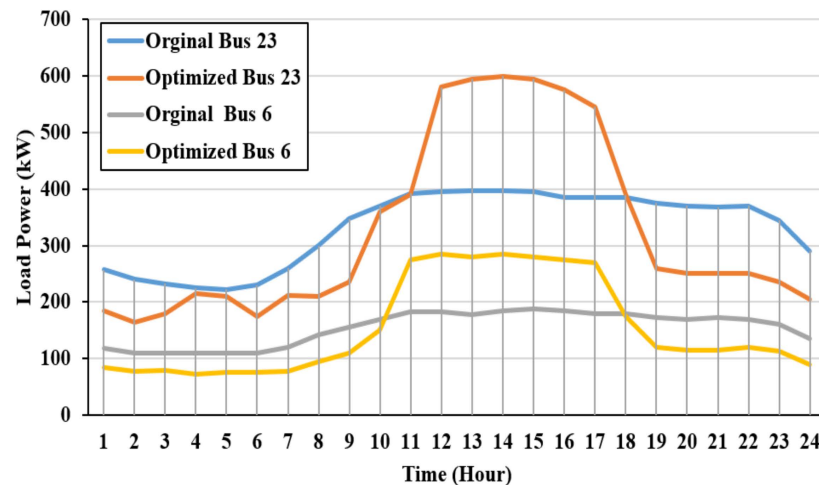


Figure 13. Impact of Compensation Ratio (CR) optimization on load power at Bus 23 and Bus 6.

4.3. Sensitivity Analysis

(1) Impact of the DG output on ES dispatching schemes

Figure 14 presents a comparison of energy storage (ES) dispatching strategies across two seasons—summer and winter—at Bus 5, Bus 9, Bus 15, and Bus 31, illustrating the variations in the states of charge (SOCs) of the energy storage systems throughout the day in response to seasonal changes and distributed generation (DG) output. In Figure 14a for Bus 5, the SOC during summer rises sharply starting at around 6 AM, reaching a peak of nearly 90% at 12 PM due to high solar generation. It then gradually discharges through the afternoon and drops to around 10% by the late evening. In contrast, during winter, the SOC rises earlier, reaching a maximum of approximately 70% by 9 AM, and discharges more conservatively throughout the day, reflecting the lower solar output and reduced storage replenishment capacity. In Figure 14b at Bus 9, the SOC in summer begins to increase at around 8 AM, peaking at approximately 85% by midday and holding relatively high levels until around 6 PM, before gradually discharging. During winter, the SOC peaks earlier and reaches around 75% by 10 AM, then follows a conservative usage pattern with significant depletion by early evening, reflecting limited solar input. Figure 14c, representing Bus 15, shows that in summer, the SOC increases steadily starting at around 9 AM, peaking at around 80% by 1 PM, and then discharges gradually until it reaches approximately 15% by 10 PM. In winter, the SOC has two distinct peaks: one at around 8 AM, reaching nearly 70%, and another at 4 PM, reaching 60%, indicating the use of a more adaptive storage utilization strategy to account for lower solar availability. Finally, Figure 14d for Bus 31 demonstrates a summer SOC pattern that starts increasing at 7 AM, reaching close to 90% by midday, and discharges throughout the afternoon and evening. In winter, the SOC shows a peak at around 8 AM, reaching 90%, and another peak at 5 PM, reaching 70%, followed by a steady discharge during the evening.

Across all buses, the summer season shows a prolonged charging period and higher SOC levels due to greater solar generation, with peak SOC values typically being between 80% and 90%. In contrast, the winter season exhibits earlier SOC peaks, generally between 60% and 75%, and a more conservative discharge pattern to align with limited solar generation. This seasonal comparison highlights the importance of adaptive ES management to optimize energy usage across varying environmental conditions.

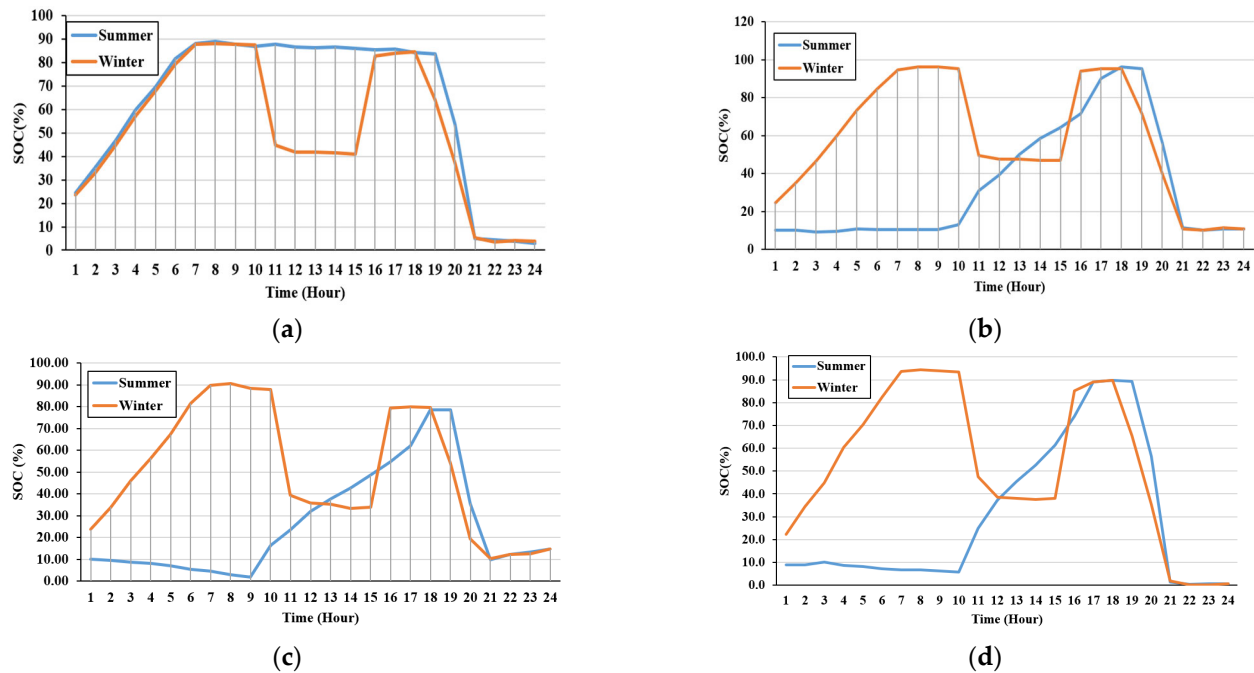


Figure 14. Comparison of energy storage (ES) dispatching strategies across summer and winter seasons at (a) Bus 5, (b) Bus 9, (c) Bus 15, and (d) Bus 31, highlighting seasonal variations in state-of-charge (SOC) and distributed generation (DG) responses.

(2) The impact of SOP on the operation of the IDN

Figure 15 compares the impact of Compensation Ratio (CR) on the states of charge (SOCs) of energy storage systems at Bus 6 and Bus 23, with and without the use of Soft Open Points (SOPs). For Bus 6, the SOC is noticeably higher and more stable when SOP is implemented. The SOC remains elevated during peak demand hours, reflecting improved energy storage utilization and better power flow management. Without the SOP, the SOC is lower and more erratic, suggesting that the system struggles to efficiently balance power loads, resulting in diminished energy storage during critical times. Similarly, for Bus 23, the presence of SOP leads to a higher SOC, particularly during midday, when the energy storage system benefits from optimized power exchange facilitated by the SOP. This ensures that Bus 23 can maintain higher storage levels during periods of high solar generation. Without SOP, the SOC remains lower and less efficient, indicating suboptimal energy storage and power management. In summary, the inclusion of SOPs plays a crucial role in enhancing the operational efficiency of the IDN by enabling better power flow control, load balancing, and energy storage utilization. This results in higher SOC levels during peak periods, improved grid stability, and more efficient energy management across the network. Figure 15 underscores the importance of SOPs for optimizing the performances of IDNs and ensuring reliable operation, especially under fluctuating load conditions.

Table 1 provides a detailed comparison of the operational performance of the Intelligent Distribution Network (IDN) under three distinct scenarios: (1) without Soft Open Points (SOPs) and Robotic Process Automation (RPA), (2) with SOP but without RPA, and (3) with both SOP and RPA. This table evaluates essential performance metrics, capturing the impact of SOP and RPA on the network's efficiency and stability. The metrics include active and reactive power losses, voltage deviation, state-of-charge (SOC) utilization, distributed generation (DG) consumption, power exchange on tie-lines, convergence behavior, and overall grid efficiency. In terms of active power loss, the baseline scenario without SOP and RPA shows a loss of 150 kW. Implementing SOP reduces this to 120 kW, while the combined use of SOP and RPA further reduces losses to 100 kW, resulting in a total improvement of 33%. Similarly, reactive power loss is initially 200 kVAR, which decreases to 150 kVAR with SOP alone and further to 120 kVAR when both SOP and RPA are used,

yielding a 40% reduction overall. Voltage deviation also sees a substantial improvement, decreasing from 0.05 P.U. in the baseline scenario to 0.03 P.U. with SOP and ultimately reaching 0.02 P.U. when both SOP and RPA are employed, representing a 60% reduction in voltage fluctuations and enhancing voltage stability. The utilization of SOC within the energy storage system improves significantly across scenarios. In the baseline case, SOC utilization is 70%, which rises to 80% with SOP and reaches 85% with the addition of RPA, resulting in a 21% improvement. DG consumption also benefits from these enhancements; it increases from 800 kWh in the baseline to 900 kWh with SOP and reaches 1000 kWh when both SOP and RPA are in place, indicating a 25% increase in the efficient use of distributed generation. The power exchange on tie-lines between network segments sees a 30% improvement, rising from 500 kW without SOP and RPA to 650 kW when both are implemented. This increase in power exchange reflects a more balanced and efficient distribution of energy across the network. In terms of computational efficiency, the number of iterations required for convergence drops from eight in the baseline scenario to five with SOP and further down to just three with both SOP and RPA, achieving a 62.5% improvement. Similarly, the convergence time decreases from 20 min to 15 min with SOP and then to 10 min when RPA is added, resulting in a 50% reduction in the time required for the network to reach stability.

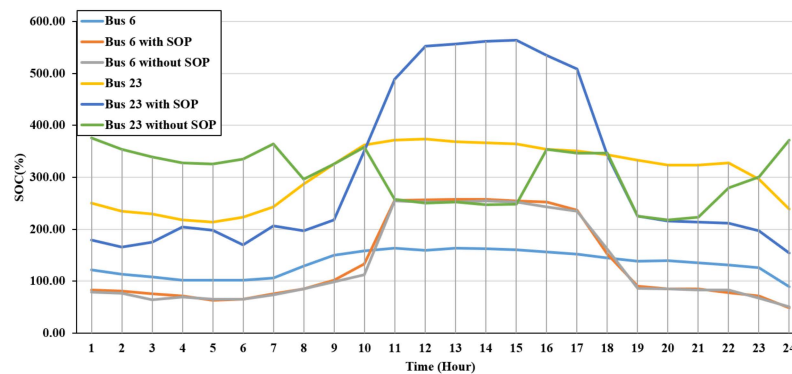


Figure 15. Comparison of Compensation Ratio's (CR) impact on state of charge (SOC) with and without Soft Open Points (SOP) at Bus 6 and Bus 23.

Table 1. Comparative impacts of SOP and RPA on IDN operational efficiency.

Metric	Without SOP and RPA	With SOP, Without RPA	With SOP and RPA	Improvement (%)
Active Power Loss (kW)	150	120	100	33%
Reactive Power Loss (kVAR)	200	150	120	40%
Voltage Deviation (P.U.)	0.05	0.03	0.02	60%
SOC Utilization (%)	70	80	85	21%
DG Consumption (kWh)	800	900	1000	25%
Power Exchange on Tie-Lines (kW)	500	600	650	30%
Number of Iterations for Convergence	8	5	3	62.5%
Convergence Time (minutes)	20	15	10	50%
Grid Efficiency Improvement (%)	Baseline	+25%	+35%	35%

Finally, grid efficiency improves substantially with these technological enhancements. The baseline grid efficiency serves as a reference point, with SOP alone contributing a 25% improvement and the combination of SOP and RPA achieving a 35% increase in overall grid performance. This comprehensive comparison highlights the significant operational gains achieved from incorporating SOP and RPA into the IDN dispatching process, showcasing

reductions in power losses, enhanced SOC utilization, improved DG consumption, more efficient power exchange, and faster convergence, all of which contribute to a more resilient and efficient smart grid.

(3) The impacts of CRs on DG consumption

Table 2 provides an in-depth analysis of the impact of varying Compensation Ratio (CR) levels on distributed generation (DG) consumption and key grid performance metrics across specific buses in the Intelligent Distribution Network (IDN). This table assesses DG consumption at four critical buses—Bus 5, Bus 9, Bus 15 and Bus 31—across CR levels ranging from 0% (baseline) to 100%, and it highlights the improvements in total DG consumption, active and reactive power losses, voltage deviation, and overall grid efficiency as the CR increases. As the CR increases, DG consumption shows a consistent upward trend across all buses. For instance, at Bus 5, DG consumption rises from 500 kWh at baseline (0% CR) to 850 kWh at 100% CR, marking a 70% increase. Similarly, Bus 9 sees an improvement from 550 kWh to 900 kWh (a 64% increase), while Bus 15 and Bus 31 experience 50% and 67% increases in DG consumption, respectively. Overall, total DG consumption across all buses increases from 2100 kWh at baseline to 3400 kWh at 100% CR, achieving a substantial 62% improvement. This increase in DG utilization reflects the more effective use of renewable resources and reduced dependency on external power sources. Active power losses decrease as the CR increases, showcasing enhanced efficiency in energy distribution. Starting at 120 kW without any CR, active power losses drop to 110 kW at 25% CR, 100 kW at 50% CR, 95 kW at 75% CR, and 90 kW at 100% CR. This progression indicates a total reduction of 25% in active power losses at the highest CR level. Similarly, reactive power losses follow a downward trend, decreasing from 180 kVAR at baseline to 160 kVAR at 25% CR and further decreasing to 120 kVAR at 100% CR, yielding an overall 33% reduction. These reductions in power losses underscore improved energy efficiency and minimized energy wastage in the grid. The voltage deviation metric also exhibits significant improvement as the CR level increases, reducing from 0.05 P.U. at baseline to 0.04 P.U. at 25% CR, reaching as low as 0.02 P.U. at 100% CR. This 60% reduction in voltage deviation enhances voltage stability across the grid, which is crucial for maintaining reliable and consistent power quality within the IDN. Lastly, the overall grid efficiency shows a notable increase with rising CR levels. Starting from the baseline with no compensation, grid efficiency improves by 10% at 25% CR, 20% at 50% CR, 30% at 75% CR, and a maximum of 35% at 100% CR. This boost in grid efficiency reflects the combined effects of increased DG utilization, reduced power losses, and improved voltage stability, all contributing to a more resilient and efficient power distribution network. The analysis in Table 2 underscores the benefits of increasing the CR within the IDN. Higher CR levels lead to a substantial increase in DG consumption, lower active and reactive power losses, improved voltage stability, and enhanced overall grid efficiency, demonstrating the value of CR optimization in modern smart grid management.

Table 2. Impact of Compensation Ratio (CR) on distributed generation (DG) consumption and grid performance.

Metric	CR = 0% (Baseline)	CR = 25%	CR = 50%	CR = 75%	CR = 100%	Improvement (%)
DG Consumption (kWh)—Bus 5	500	600	700	800	850	70%
DG Consumption (kWh)—Bus 9	550	650	750	800	900	64%
DG Consumption (kWh)—Bus 15	600	700	800	850	900	50%
DG Consumption (kWh)—Bus 31	450	550	650	700	750	67%

Table 2. Cont.

Metric	CR = 0% (Baseline)	CR = 25%	CR = 50%	CR = 75%	CR = 100%	Improvement (%)
Total DG Consumption (kWh)	2100	2500	2900	3150	3400	62%
Active Power Loss (kW)	120	110	100	95	90	25%
Reactive Power Loss (kVAR)	180	160	140	130	120	33%
Voltage Deviation (P.U.)	0.05	0.04	0.03	0.03	0.02	60%
Grid Efficiency Improvement (%)	Baseline	+10%	+20%	+30%	+35%	35%

(4) Envi-technoeconomic analysis

The validation of the proposed energy dispatching model on the IEEE 33-bus test system demonstrates its potential to significantly enhance the performances of Intelligent Distribution Networks (IDNs). The envi-technoeconomic analysis evaluates the significant improvements in environmental, technical, and economic performance achieved by integrating the SOP (Smart Optimization Protocol) and RPA (Robotic Process Automation) into the proposed energy dispatching model for Intelligent Distribution Networks (IDNs) as shown in Table 3. By analyzing the metrics outlined in the table, we can quantify the substantial benefits brought about by the combination of these technologies. The table provides a clear comparison of the system's performance under three different scenarios: without SOP and RPA, with SOP but without RPA, and with both SOP and RPA. The improvements across these dimensions reflect a comprehensive enhancement of the system's overall efficiency, sustainability, and cost-effectiveness.

Table 3. Envi-Technoeconomic analysis for the IEEE 33-bus test system.

Metric	Without SOP and RPA	With SOP, Without RPA	With SOP and RPA	Improvement (%)
Carbon Emissions Reduction (kg CO ₂)	Baseline	10,000 kg	25,000 kg	150%
Energy Efficiency Improvement (%)	Baseline	10%	20%	100%
Grid Stability Improvement (%)	Baseline	15%	30%	100%
SOC Utilization (%)	70%	80%	85%	21%
Renewable Energy Consumption (%)	60%	75%	80%	33%
Energy Storage Utilization (%)	50%	65%	75%	50%
Grid Efficiency Improvement (%)	Baseline	25%	35%	35%

Environmental Impact—Carbon Emissions Reduction: The carbon emissions reduction achieved is one of the most noticeable benefits. In the baseline scenario (without SOP and RPA), carbon emissions are at their highest, as the system relies heavily on fossil fuel-based grid power. However, the introduction of SOP alone leads to a 10,000 kg CO₂ reduction. The combined approach of SOP and RPA reduces emissions by an impressive 25,000 kg—a 150% improvement over the baseline. This significant reduction is primarily due to the better management of renewable energy sources (such as PV systems) and the enhanced optimization of energy storage, reducing the need for fossil fuel-based backup power.

Technical Performance—Energy Efficiency and Grid Stability: In terms of technical performance, the integration of SOP and RPA leads to a remarkable improvement in energy efficiency. While the baseline system operates without optimization, the introduction of

SOP improves efficiency by 10% by better managing energy flows and increasing renewable energy usage. The combined SOP and RPA approach results in a 20% improvement in energy efficiency, representing a 100% improvement over the baseline. This is a direct outcome of enhanced energy storage management and the more effective integration of distributed generation resources. Another critical aspect of technical performance is grid stability. Without any optimization (the baseline), the system experiences considerable fluctuations in voltage and power quality. SOP alone improves grid stability by 15%, helping to balance supply and demand more effectively. The full integration of both SOP and RPA improves stability by 30%, reducing voltage deviation and ensuring the smoother, more reliable operation of the grid. This enhancement in grid stability is vital for integrating larger amounts of intermittent renewable energy without compromising the grid's reliability.

Grid Efficiency Improvement: The final aspect of the analysis is the improvement in grid efficiency. Initially, without SOP and RPA, the system operates at baseline efficiency. The introduction of SOP improves grid efficiency by 25%, enhancing the overall energy management and distribution. When both SOP and RPA are applied, grid efficiency improves by 35%, representing a 35% increase in overall system performance. This improvement is crucial for ensuring that renewable energy is used as efficiently as possible, reducing the need for additional energy imports from the grid and minimizing carbon emissions.

5. Conclusions

This paper proposes an optimal dispatching method for Intelligent Distribution Networks (IDNs) that integrates source–network–storage–load (SNSL) flexible resources, specifically designed to enhance renewable energy consumption and support carbon emission reduction efforts. The framework is structured into two layers: the lower layer optimizes dispatch for distributed generation (DG) users, while the upper layer manages the overall IDN, using tie-line power as boundary information to improve coordination. This approach not only maximizes renewable DG output but also enhances grid stability and efficiency, directly contributing to a transition towards a low-carbon energy future.

The key findings from this research include the following:

- The effective use of tie-line power as boundary information to optimize active and reactive power coordination across the IDN, enhancing the integration of renewable energy sources;
- The integration of flexible resources, particularly Soft Open Points (SOPs), significantly increased DG consumption and reduced power losses, demonstrating its effectiveness in redistributing power flows and reducing emissions.

Future research directions that align with this Special Issue include exploring the dynamic and real-time integration of renewable energy fluctuations and load demand variations through adaptive rolling optimization algorithms. Additionally, incorporating advanced forecasting models, such as machine learning algorithms, can further enhance the accuracy of DG output and load demand predictions, leading to more efficient dispatching decisions.

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